



Uncertainty Propagation and Sensitivity Analysis in Additive Manufacturing Simulation

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- ▶ Model & Methodology
- ▶ Construction of probabilistic model
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Context and Objective

- ▶ Safran Additive Manufacturing Campus collaborated with Phimeca.
- ▶ Goal: Develop an uncertainty propagation chain for additive manufacturing.
- ▶ Objective: Identify the most influential parameters to better control the process.

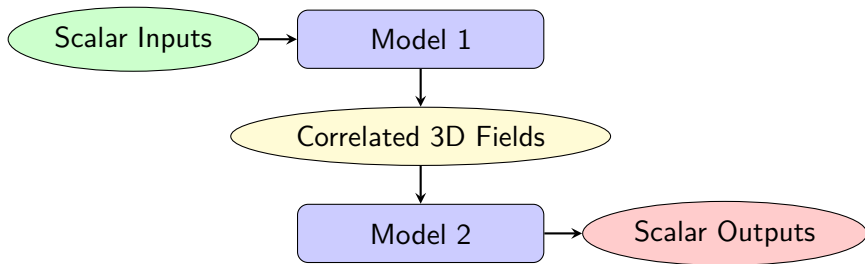
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Hybrid Simulation Model

Two intermediate models

- ▶ **Model 1:** Inputs are scalar values, outputs are correlated 3D fields.
- ▶ **Model 2:** Inputs are 3D fields, outputs are scalar parameters.



Methodology and Tools

Methodology

- ▶ Construction of probabilistic models for uncertain inputs.
- ▶ Design and computation of a Design of Experiments (DoE).
- ▶ Development of a Kriging metamodel for Model 1.
- ▶ Sobol sensitivity analysis applied to the Kriging metamodel.
- ▶ Approximation of Model 2 using PCA-PCE metamodel.
- ▶ Both Sobol and Shapley sensitivity analyses applied to the PCA-PCE metamodel.

Tools

- ▶ OpenTURNS, OtShapley & Persalys
- ▶ Other Python packages : Scikit-Learn & Scipy

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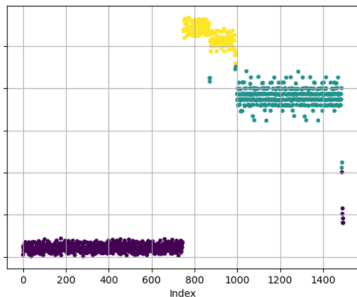
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Construction of probabilistic model

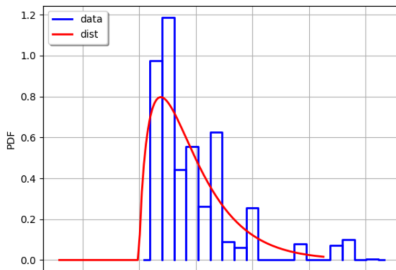
- ▶ A total of 13 scalar parameters considered as uncertain inputs for model 1.
- ▶ Existing data available for some parameters.
- ▶ For variables with available data, statistical inference is performed.
- ▶ For other variables, statistical distributions are constructed based on available statistical information such as minimum/maximum values and mode.
- ▶ In such cases, the parameters of the proposed distributions are optimized to match the provided statistical characteristics.
- ▶ Some datasets contain only final values, without distinction between nominal value and local variability.
- ▶ To address this, a clustering analysis is applied to identify data subgroups and extract the mean and standard deviation for each cluster.

Construction of probabilistic models

Example of clustering



Example of fitted distributions



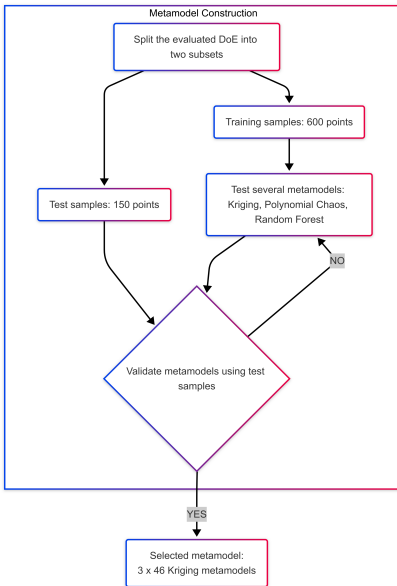
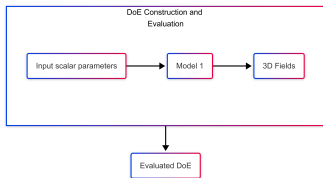
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Metamodeling & Sensitivity analysis (SA)

- ▶ Construct a Latin Hypercube Sampling (LHS) Design of Experiments (DoE) with a total of 750 points.
- ▶ Split the dataset into 600 training samples and 150 test samples.
- ▶ Fit 3×46 Kriging metamodels on the training set (3 fields indexed over 46 spatial locations)
- ▶ Perform sensitivity analysis using Sobol indices, based on the constructed probabilistic model.

Construction of DoE & Metamodels



Sensitivity Analysis with Sobol Indices (SI)

Methodology

- ▶ Performing Sobol analysis on each output.
- ▶ 3 x 3D fields indexed on 46 (dim 1) x 10 (dim 2) x 15 (dim 3) special locations (total of 6900 nodes/field).
- ▶ The SI are aggregated, for each field, and for each couple of dimensions.

Parameters

- ▶ $N_{mc} = 150000 \rightarrow N_{total} = 15000 * (d + 2)$ where $d = 13$
- ▶ Estimator: **Mauntz-Kucherenko**.

Aggregated Sobol Index

$$S_{i,\text{agg}} = \frac{\sum_{k=1}^d S_{ik} \cdot V_k}{\sum_{k=1}^d V_k}$$

i : input variable

k : output component

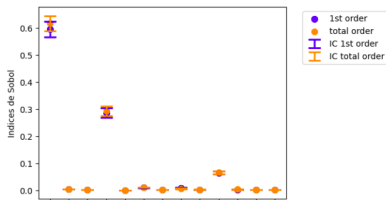
d : total number of outputs

S_{ik} : Sobol index for i and k

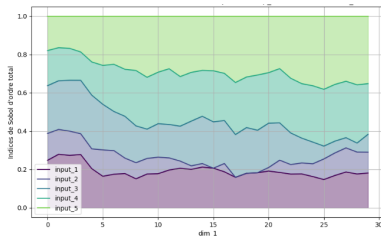
V_k : variance of output k

Sobol Results

Aggregated SI across all dimensions for Field 1



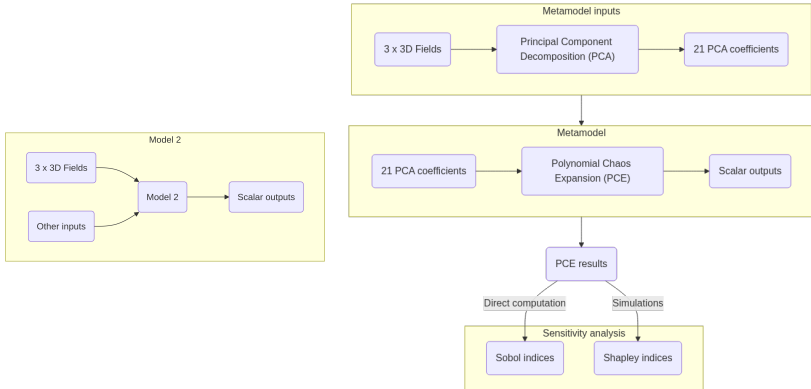
Aggregated SI across Dimensions 2 & 3 for Field 1



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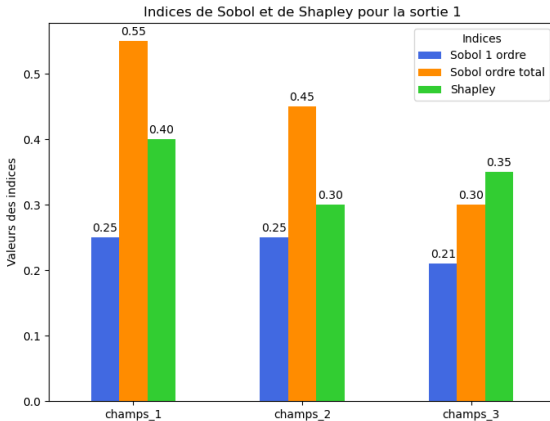
Model 2: Metamodel & Sensitivity Analysis



Applied methodology

- ▶ Generate 1000 LHS samples from input distributions.
- ▶ Split into 800 training and 200 test samples.
- ▶ Evaluate Model 1 (kriging) to produce 3D fields.
- ▶ Use fields as inputs to Model 2 to get scalar outputs.
- ▶ Apply PCA (99% variance retained) → 21 coefficients.
- ▶ Infer distributions of PCA coefficients.
- ▶ Fit PCE metamodel on training samples.
- ▶ Validate PCE on test samples.
- ▶ Compute Sobol indices from PCE.
- ▶ Compute Shapley indices via simulations.
- ▶ Aggregate indices by original 3D field.
- ▶ Analyze aggregated sensitivity results.

Model 2: Sensitivity analysis results



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Conclusion

In this work, a complete uncertainty propagation chain was developed for a hybrid additive manufacturing simulation model.

This chain includes:

- ▶ Probabilistic modeling of input parameters
- ▶ Design of experiments (DoE) and model evaluations
- ▶ Construction of metamodels
- ▶ Sensitivity analysis of outputs with respect to inputs

Several techniques were employed:

- ▶ Clustering (K-Mean), Latin Hypercube Sampling (LHS), Kriging, Polynomial Chaos Expansion (PCE),
- ▶ Principal Component Analysis (PCA), Shapley & Sobol indices

Multiple tools were used, mainly:

- ▶ **OpenTURNS**, **OtShapley**, and other Python packages



Thank you for your attention!

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